EXPERT OPINION

1. Introduction
2. Patients and methods
3. Results
4. Discussion
5. Conclusion

Effect of imidapril versus ramipril on urinary albumin excretion in hypertensive patients with type 2 diabetes and microalbuminuria

Roberto Fogari, Amedeo Mugellini, Annalisa Zoppi, Roberta Gualtierotti, Pierangelo Lazzari, Giuseppe Derosa & Angela D'Angelo
University of Pavia, Centro per l'Ipertensione e la Fisiopatologia Cardiovascolare, Department of Internal Medicine and Therapeutics, Pavia, Italy

Objective: Aim of this study was to compare the antiproteinuric effect of imidapril (I) and ramipril (R) in diabetic hypertensive patients with microalbuminuria.
Research design and methods: One hundred and seventy-six patients were randomised to I 10 – 20 mg once daily (od) (n = 88) or R 5 – 10 mg od (n = 88) for 24 weeks. Clinic, ambulatory, central blood pressure (BP), urinary albumin excretion (UAE), plasma Angiotensin II (Ang II), bradykinin and brain natriuretic peptide (BNP) were assessed at baseline and after 6, 12 and 24 weeks.
Results: Both I and R produced a similar decrease in clinic, ambulatory and central BP ($p < 0.001$ vs baseline). Both treatments significantly reduced UAE throughout the study, but the decrease in UAE associated with I was more pronounced, being evident at week 6 ($p = 0.05$) and maximal at week 24 end-point (-42 vs –29%, $p < 0.01$). BNP and Ang II levels were similarly reduced by I and R, while bradykinin increased more with R (+132 vs +86%, $p < 0.05$).
Conclusions: These findings showed that in diabetic hypertensive patients with microalbuminuria, despite equivalent BP-lowering effect, I produced a greater antiproteinuric effect than R, which might be due to different intrinsic molecular properties of the two drugs.

Keywords: diabetes, hypertension, imidapril, microalbuminuria, ramipril

Expert Opin. Pharmacother. (2013) 14(18):2463-2473

1. Introduction

The renin-angiotensin system (RAS) is known to play a pivotal role in the pathogenesis of diabetic nephropathy [1]. Therefore, RAS blockade has become the first-line strategy in hypertensive patients with both incipient and overt diabetic nephropathy [2]. Several large trials have demonstrated that both angiotensin-converting enzyme inhibitors (ACE-Is) [3-5] and angiotensin II receptor blockers [6-8] are effective in reducing urinary albumin excretion (UAE), which is considered a surrogate renal end-point even in the micro-albuminuric range (i.e., UAE of 30 – 300 mg per 24 h) [9,10]. Various ACE-Is, however, seem to differ somehow in their ability to reduce UAE, particularly in type 2 diabetic patients. Ramipril in particular has shown some limits in these patients. In this regard, Trevisan *et al.* [11] demonstrated that ramipril at the dose of 1.25 mg/day administered to both normotensive and hypertensive patients with type 2 diabetes mellitus and microalbuminuria induced only a transient reduction in UAE, which was significant as early as after 1 month

informa healthcare

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

10.1517/14656566.2013.848195 © 2013 Informa UK, Ltd. ISSN 1465-6566, e-ISSN 1744-7666

All rights reserved: reproduction in whole or in part not permitted
RIGHTSLINK

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

of administration but did not persist thereafter, so that at the end of the 6-month study microalbuminuria was not different from baseline. This study had the main limitations in the selected population, which included both normotensive and hypertensive diabetics and in the low dose of ramipril. The limited efficacy of ramipril in reducing microalbuminuria in type 2 diabetic patients was successively confirmed by Schnack *et al.* [12] that did not observe any significant decrease in UAE after 1-year treatment with ramipril administered at the dose of 2.5 – 5 mg/day to 105 hypertensive patients with type 2 diabetes and microalbuminuria. Also, the Noninsulin-Dependent Diabetes, Hypertension, Microalbuminuria, Proteinuria, Cardiovascular Events and Ramipril trial [13], conducted in proteinuric and microalbuminuric type 2 diabetic patients, has concluded that a low daily dose of ramipril (1.25 mg) had no statistically significant beneficial renal effects over placebo after 4 years of follow-up.

Based on the available data, the ACE-I imidapril seems to have a marked antiproteinuric effect, somewhat independent of its blood pressure (BP)-lowering effect. Thus, in hypertensive patients with diabetic nephropathy Oda *et al.* [14] observed that treatment with imidapril 5 mg/die was associated with a progressive decrease in UAE, which reached statistical significance after 12 weeks and persisted during the treatment period with a reduction of 27% after 24 weeks. This study had the main limitation in the dishomogeneity of the selected population, which included both micro- and macroalbuminuric patients. Similar data, however, have been reported by Umeda [15] that in hypertensive diabetics with microalbuminuria observed that imidapril at the dose of 2.5 – 5 mg/day produced a marked reduction in UAE, which was statistically significant as early as after 4 weeks of treatment and persisted with even lower values after 12 weeks.

Thus, in the treatment of hypertensive type 2 diabetics with microalbuminuria imidapril seems to produce a greater reduction in UAE than ramipril with a better time course of such antiproteinuric effect. However, to date, there have been no head-to-head comparative studies examining the effect of these two ACE-Is in hypertensive patients with type 2 diabetes and microalbuminuria. Given this background, the present study was undertaken to compare the antiproteinuric effect of imidapril and ramipril in hypertensive patients with type 2 diabetes and microalbuminuria.

2. Patients and methods

This was a prospective, randomised, open-label, blinded endpoint parallel-group study. After discontinuation of previous antihypertensive treatment, if any, male and female outpatients aged 25 – 75 years with mild to moderate essential hypertension [(defined as sitting systolic blood pressure (SBP) ≥ 140 and < 180 mmHg and sitting diastolic blood pressure (DBP) ≥ 90 and < 105 mmHg)], type 2 diabetes mellitus well controlled by diet or oral hypoglycemic agents [glycated hemoglobin (HbA_{1c}) < 7%, absence of glycosuria and no change in hypoglycemic drugs in the last 6 months] and in the higher range of microalbuminuria defined as UAE > 150 and < 300 mg/24 h in two distinct 24-h urine collections entered a wash-out run-in period for 2 weeks. At the end of this period, the patients who fulfilled the inclusion criteria were randomised to receive imidapril 10 mg once daily (od) or ramipril 5 mg od for 24 weeks (**Figure 1**). After 2 weeks therapy, if clinic BP was > 130/80 mmHg, the dose was doubled. After 4 weeks, if BP was yet > 130/80 mmHg hydrochlorothiazide 12.5 mg od was added and after 6 weeks, if BP was yet uncontrolled, it was doubled to 25 mg daily. At week 8 the patients with BP > 130/80 mmHg were withdrawn.

Exclusion criteria included secondary hypertension, myocardial infarction or stroke within 12 months before the beginning of the study, congestive heart failure, angina, clinically significant arrhythmias, pregnancy, lactation, renal or hepatic dysfunction, cancer or any severe disease likely to interfere with the conduct of the study. Patients with known or suspected intolerance to ACE-Is were likewise excluded.

The study protocol was approved by the local Ethics Committee and informed consent was obtained from each patient before enrolment.

Clinic BP, ambulatory blood pressure (ABP), central BP, UAE, serum creatinine and potassium were evaluated at the end of the wash-out run-in period (baseline), after 4, 6, 12 and 24 weeks of treatment and after 2 weeks of treatment withdrawal. Plasma Angiotensin II (Ang II), bradykinin and brain natriuretic peptide (BNP) were also evaluated at the end of the wash-out period and after 24 weeks of treatment. Clinic BP was measured in the morning, immediately before the daily dose of medication (i.e., at trough, 24 h after dosing) by the same clinician, using the same calibrated mercury sphygmomanometer (Korotkoff I and V) on the same patients in the dominant arm, after the subject had been sitting for 5 min in a quiet room. Three BP measurements were made with a 2-min interval between them and averaged. ABP monitoring was performed over 24 h using a clinically validated device (Spacelabs 90207, Spacelabs Inc., Redmond, Wag) [16], which was programmed to measure BP every 15 min during the entire course of recording. Each recording was started in the morning, immediately after clinic BP assessment and drug administration. Patients were instructed to remain motionless each time a reading was taken. Analysis of 24-h BP recordings was preceded by removal of artefacts, according to previously described editing criteria [16,17]. Recordings were considered valid when no more than 2 nonconsecutive hours were missing over 24 h. For each patient, the following data related to SBP, DBP and heart rate (HR) were obtained through analysis of the recordings: 24-h mean values, day-time (07.00 am – 11.00 pm) and night-time (11.00 pm – 07.00 am) mean values.

Central BP was undertaken at each visit by radial artery applanation tonometry (Sphygmocor; Medical Pty Ltd. Sydney, Australia). The Sphygmocor algorithm recreates the pulse wave at the aortic root and is used to calculate the central aortic pressure [18].

RIGHTSLINK

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

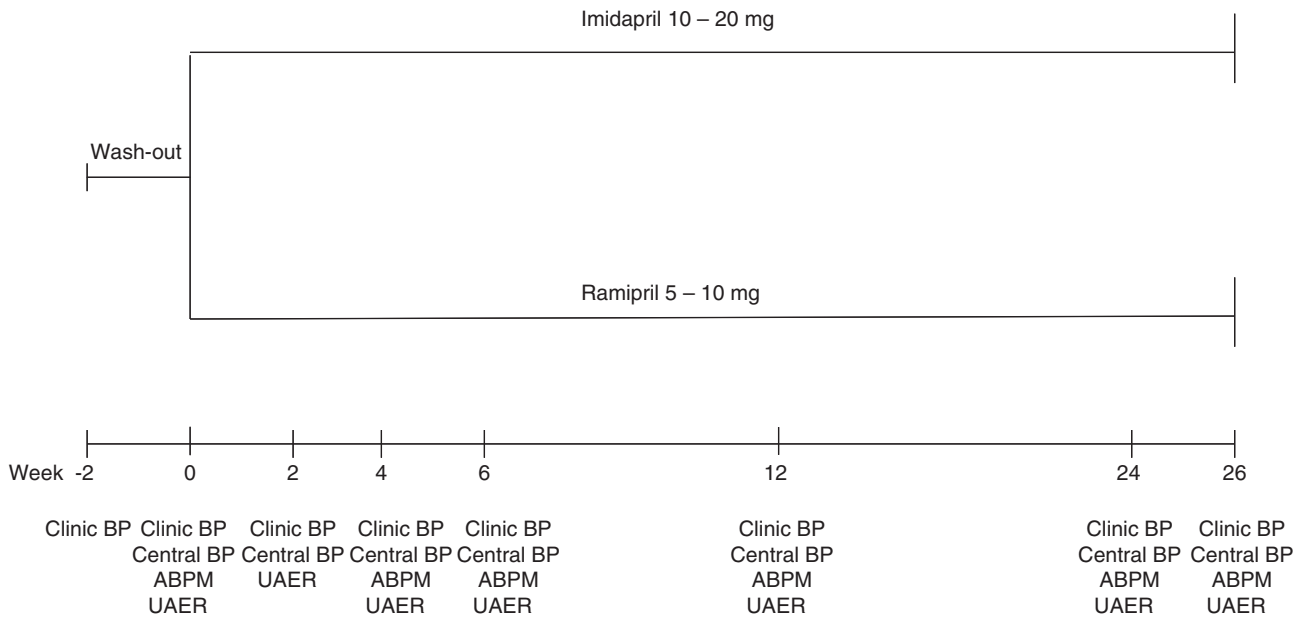

Figure 1. Study design.
ABPM: Ambulatory blood pressure monitoring; BP: Blood pressure; UAER: Urinary albumin excretion rate.

UAE was assessed by means of radioimmunoassay; each reported value represents the mean of two distinct 24-h urine collections within two consecutive days.

Blood samples for creatinine, potassium, Ang II, bradykinin and BNP were taken always at the same hour in the morning and after centrifugation plasma was frozen (-80°C). Serum creatinine and potassium were measured using commercial kits employing routinely used clinical chemistry procedures.

Blood samples for the determination of plasma Ang II concentration were drawn from patients in supine position after 1 h of complete rest and were collected into prechilled 10-ml syringes prepared with 125 mmol EDTA and 26 mmol phenanthrolin (Merck KGaA, Darmstadt, Germany) to inhibit ACE. The samples were centrifuged for 10 min at 4°C immediately after collection and plasma was stored rapidly after centrifugation at -21°C and analysed within 3 months. Plasma samples (1 ml) were extracted with Bond Elut PH cartridges (PK 100, ICT-ASS-Chem, Analytichem, Harbor City, CA, USA) and bound angiotensin was eluted with 0.5 ml methanol. Subsequently, plasma extracts were evaporated in a Speed-Vac (SVC 100, Savant Instruments, Pvt Ltd, Hyderabad, India) for 1 h under reduced pressure. Immediately after purification of the samples, immunoreactive Ang II was measured by radioimmunoassay with antiserum and labelled Ang II (NEX 105, Du Pont) [19]. Cross-reactivity of this method is 100% for Ang II and 1.2% for Ang I. The quantification of radioactivity was performed with a gamma counter. Measurements made by this method are accurate within a range from 1.9 to 32 pg Ang II ml^{-1}. All determinations of immunoreactive Ang II were made in duplicate and the mean values are given. The coefficient of variation was 5%.

Blood sample for measurements of bradykinin were drawn into a cold anhydrous ethanol (4:1 blood to ethanol). After 1 h at 4°C the mixture was centrifuged (15 min, 3000 rpm, 10°C) and the supernatant was decanted and frozen at -80°C until assay. Bradykinin concentration was determined using a commercially available enzyme immunoassay (Peninsula Laboratories, Division of Bachem, San carlos, CA, USA) [20].

For measurements of BNP a test tube containing EDTA-2Na was used. Plasma was separated by centrifugation and stored at -80°C. Thereafter the plasma BNP concentrations were determined by immunofluorescence (Axsym, Abbot Laboratories). The intraassay variation was 4.9 – 5.4%. The lower detection limit was 6 pg/ml. Concentration of BNP below the detection limit was assigned a value of 5 pg/ml [21].

At each visit, adverse events (AEs) spontaneously reported by patients or elicited by the investigators were recorded.

2.1 Study end-points

The primary end-point was change in UAE after 24 weeks of treatment with imidapril and ramipril. Secondary end-points included changes from baseline in clinic, ambulatory and central SBP and DBP values. The following variables were also included to determine the impact of treatment on selected biomarkers, e.g., changes in Ang II, bradykinin and BNP levels.

2.2 Sample size

The population so far studied was very heterogeneous, with a great variability in proteinuria. After a log-transformation of the data, we assume that the mean of ln (albuminuria) in

RIGHTSLINK

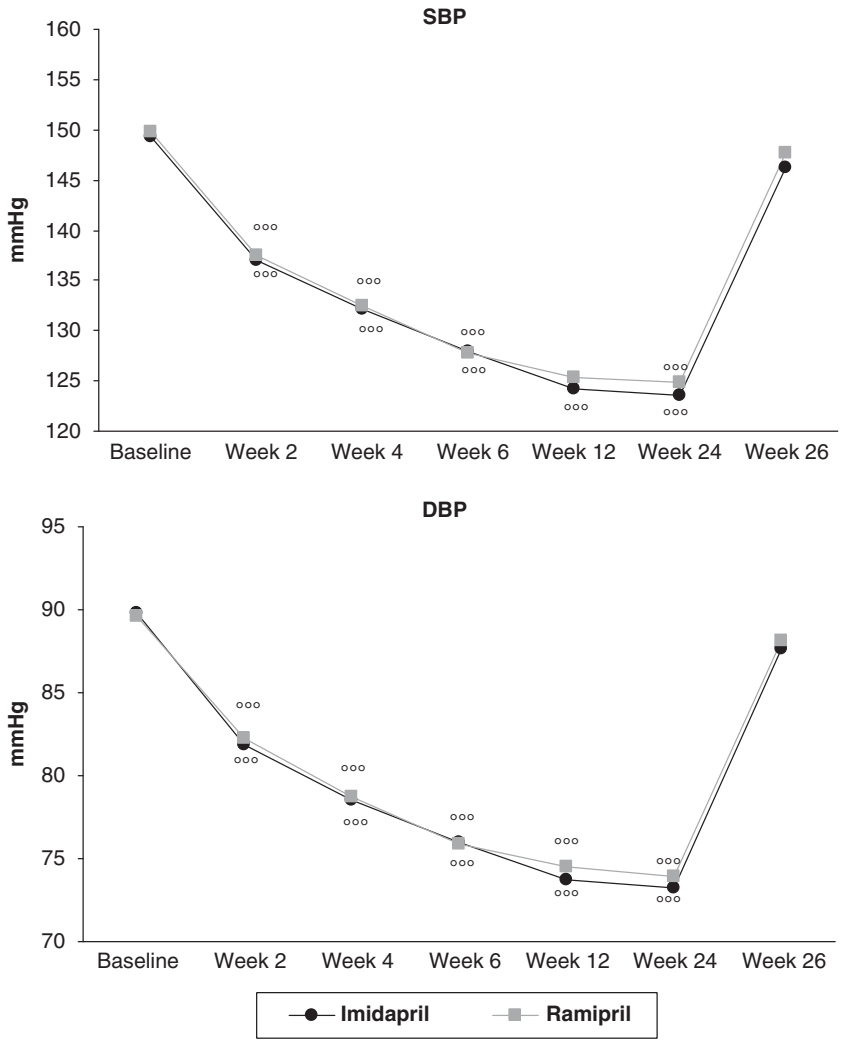

Figure 2. Clinic blood pressure.
°°°p < 0.001 vs baseline.
BP: Blood pressure; DBP: Diastolic blood pressure; SBP: Systolic blood pressure.

this study population is 5.42 mg/24 h (considering that only patients with albuminuria in the range 150 – 300 mg/24 h are included in the study) and the standard deviation is 1.3 mg/24 h. We also assume a very low effect of ramipril (10% albuminuria reduction) and an effect of imidapril of at least 20% in this specific study population. On the log scale, this means a difference in means of about 0.5 mg/24 h. A sample size of 88 patients per group will have an 80% power to detect a difference in means of 0.5 mg (10% difference) with a standard deviation of 0.27 mg/24 h, using a two-group t-test with a 0.05 two-sided significant level.

2.3 Statistical analysis

Data are expressed as means ± SD. The homogeneity check of patient distribution between treatment groups was performed using the χ^2 test. Treatment comparison between the patients who received imidapril and those who received ramipril was performed with a two-sided t test with a significant level of 0.05. The changes in primary variable, UAE, as well as in BP were analysed statistically using analysis of covariance.

3. Results

In total, 260 consecutive outpatients were screened. Out of them, 133 patients were untreated for hypertension, 91 were treated with antihypertensive monotherapy, 30 were treated with dual combination and 6 with triple combination antihypertensive therapy. At the end of the run-in period, 211 patients (81.2%) fulfilling the inclusion/exclusion criteria were randomised to receive imidapril 10 mg or ramipril 5 mg. Thirty-five patients dropped out after randomisation (18 in the imidapril and 17 in the ramipril group, with similar characteristics), 24 patients because of uncontrolled BP (13 with

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

RIGHTSLINK

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

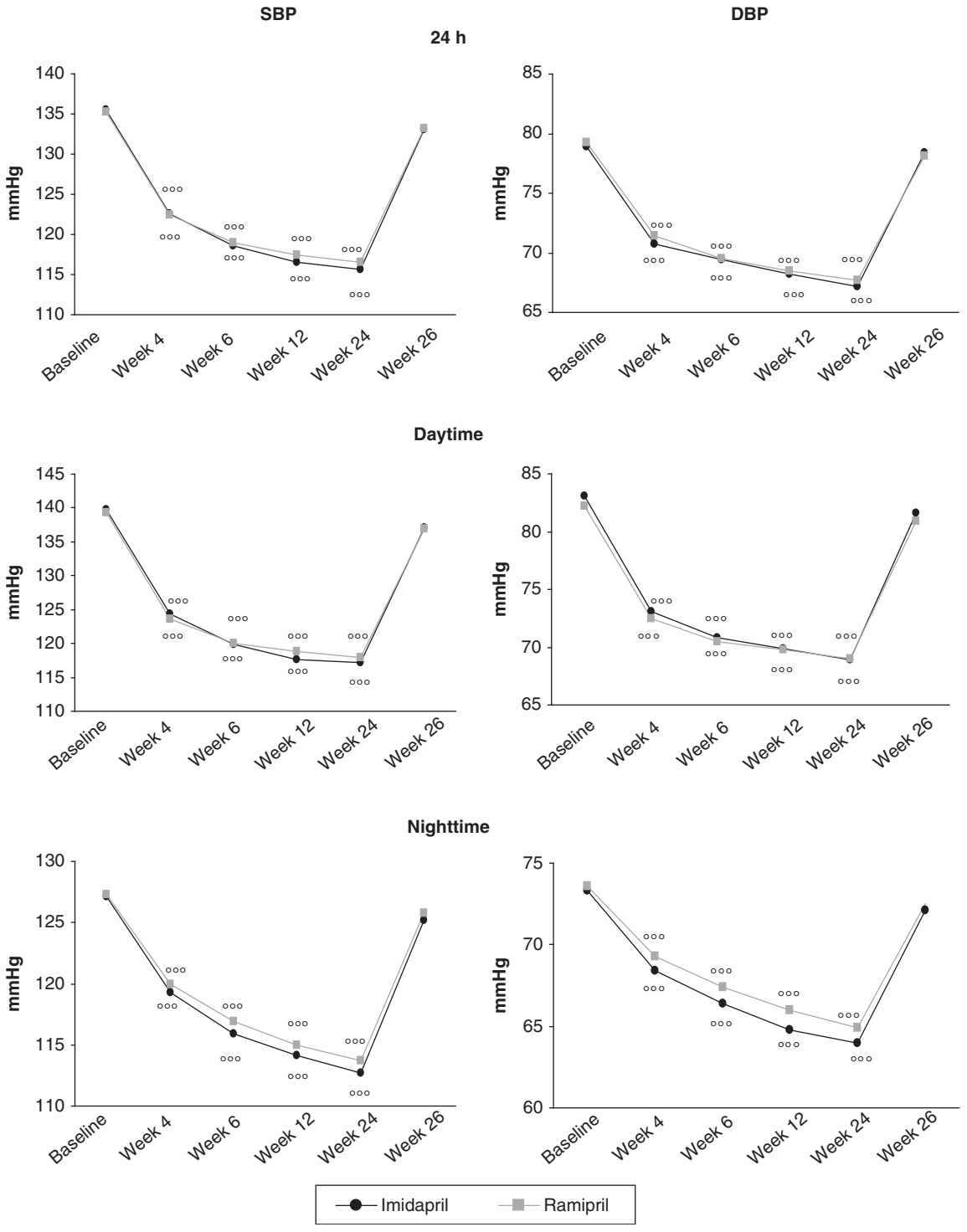

Figure 3. ABPM 24 h, day time and night time.

°°°p < 0.001 vs baseline.

DBP: Diastolic blood pressure; SBP: Systolic blood pressure.

imidapril and 11 with ramipril), 8 patients due to AEs (3 with imidapril and 5 with ramipril), 1 because of consent withdrawn (in the imidapril group) and 2 were lost at follow-up (1 in the imidapril and 1 in the ramipril group). A total of 176 patients, 88 in the imidapril group (43 males and 45 females, aged 61.1 ± 6.7 years) and 88 in the ramipril group (41 males and 47 females, aged 60.5 ± 6.4 years), completed the study and the results refer to these patients, whose

RIGHTSLINK

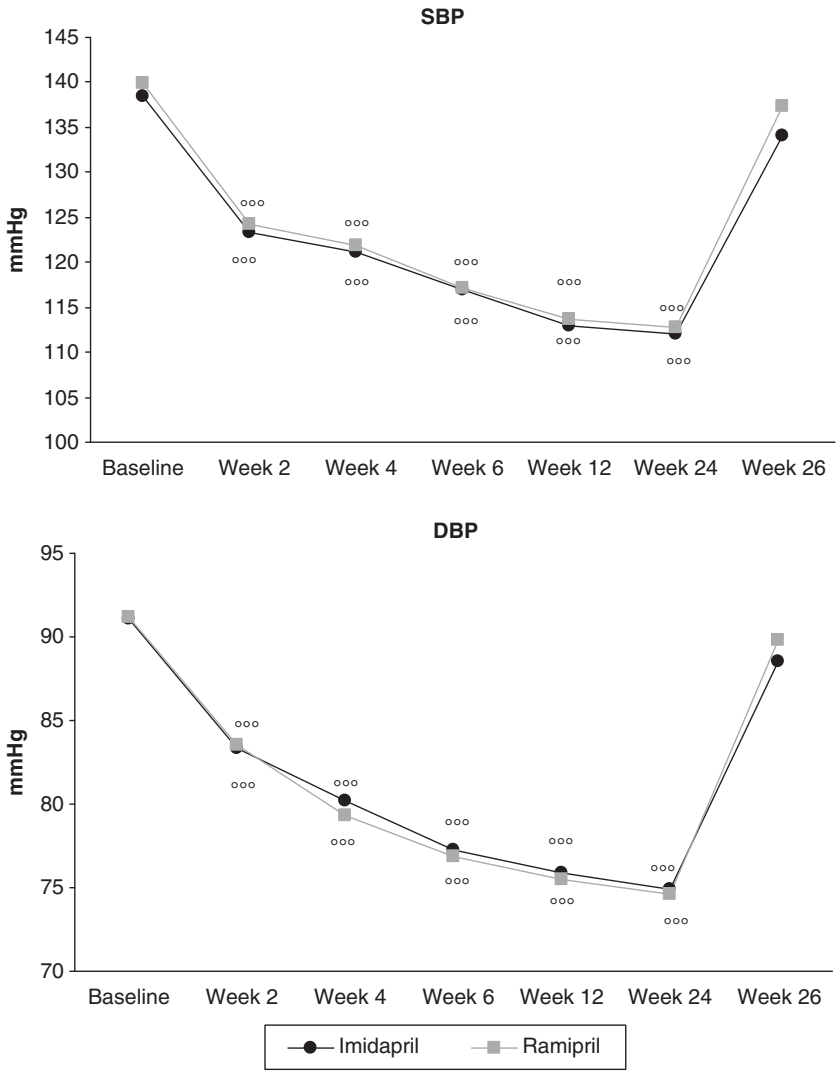

Figure 4. Central blood pressure.
°°°p < 0.001 vs baseline.
BP: Blood pressure; DBP: Diastolic blood pressure; SBP: Systolic blood pressure.

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

distribution in the two treatment groups was homogeneous with regard to the main demographic and clinic characteristics. In the imidapril group, 23 patients were treated with imidapril 10 mg monotherapy for the entire duration of the study, whereas 22 patients had the dosage of imidapril monotherapy doubled to 20 mg from week 2 onwards. The imidapril-treated patients who needed the addition of HCTZ at week 4 were 43: out of them, 19 were given HCTZ 12.5 mg throughout the remaining study period, whereas 24 needed a dose titration to 25 mg from week 6 onwards. In the ramipril group, 19 patients received ramipril 5 mg monotherapy for the entire duration of the study, whereas 24 patients had the dosage of ramipril monotherapy doubled to 10 mg from week 2 onwards. The ramipril-treated patients who needed the addition of HCTZ at week 4 were 45: out of them 19 patients received HCTZ 12.5 mg throughout the remaining study period, whereas 20 patients required a dose titration to 25 mg from 6 week onwards.

3.1 Blood pressure

Clinic BP results are reported in **Figure 2**. Both imidapril- and ramipril-based therapy produced a significant reduction in SBP and DBP mean values at week 24 end-point, with no significant difference between the two treatments (-25.0/15.7 mmHg with imidapril, $p < 0.001$ vs baseline and -25.8/16.6 mmHg with ramipril, $p < 0.001$ vs baseline). Two weeks after stopping treatment, clinic BP returned to baseline levels with no difference between the two treatment groups. Clinic HR values were unaffected by both treatments at any time of the study.

Ambulatory BP results are reported in **Figure 3**. Both imidapril- and ramipril-based therapy produced similar reductions in 24-h, day-time and night-time SBP and DBP mean

RIGHTSLINK

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

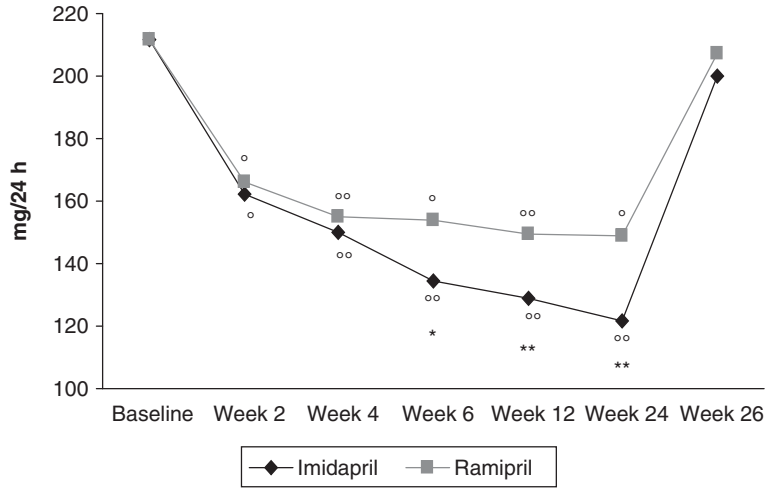

Figure 5. Microalbuminuria levels.
°p < 0.05; °°p < 0.01 vs baseline.
*p < 0.05; **p < 0.01 vs ramipril.

Table 1. BNP, bradykinin and angiotensin II levels in the imidapril- and ramipril-treated patients.

	Imidapril (n = 88)			Ramipril (n = 88)		
	Baseline	Week 4	Week 24	Baseline	Week 4	Week 24
BNP (pg/ml)	39.61 ± 22.5	27.43 ± 17.5*	25.89 ± 16.7*	38.06 ± 24.0	27.54 ± 17.5*	25.68 ± 16.2*
Bradykinin (pg/ml)	22.21 ± 11.2	37.86 ± 20.7‡	41.39 ± 21.1‡	21.98 ± 11.7	40.60 ± 22.1‡	50.97 ± 24.1‡,§
Angiotensin II (pg/ml)	30.97 ± 15.8	15.71 ± 8.0‡	16.31 ± 8.0‡	30.30 ± 15.7	14.77 ± 7.2‡	22.40 ± 13.4*

Data are expressed as mean ± SD.
*p < 0.05.
‡p < 0.01 vs baseline.
§p < 0.05 vs ramipril.
BNP: Brain natriuretic peptide.

values at week 24 end-point, with no significant difference between the two treatments. The mean decrease in 24-h SBP/DBP values was of 19.8/11.7 mmHg (p < 0.001 vs baseline) with imidapril and of 18.7/11.6 mmHg (p < 0.001 vs baseline) with ramipril. Day-time SBP/DBP levels were reduced by a mean of 22.5/14.2 mmHg (p < 0.001 vs baseline) by imidapril and by a mean of 21.3/13.2 mmHg (p < 0.001 vs baseline) by ramipril. Similar reductions were observed in night-time SBP/DBP mean values (-14.4/9.3 mmHg, p < 0.001 vs baseline with imidapril and -13.6/8.7 mmHg, p < 0.001 vs baseline with ramipril). Two weeks after stopping therapy 24-h, day-time and night-time ambulatory BP returned to baseline values in both treatment groups, with no difference between them.

3.2 Central BP

Consistent with the clinical BP measures and ambulatory measures, reductions in both central SBP and central DBP from baseline were significant with both imidapril and ramipril (**Figure 4**). At week 24 end-point, the mean decrease in central SBP/DBP was 26.3/16.2 mmHg (p < 0.001 vs baseline) with imidapril and 27.1/16.5 mmHg (p < 0.001 vs baseline) with ramipril. No significant difference between the two treatments with regard to their effect on central BP was observed at any time of the study.

3.3 Urinary albumin excretion

Both imidapril- and ramipril-based treatments were associated with a significant decrease in UAE throughout the treatment period (**Figure 5**). However, the decrease in UAE associated with imidapril was more pronounced at each time of the study. The difference versus ramipril was statistically significant at week 6 and became progressively more evident, being maximal at week 24 (end of active treatment period). At this time point, UAE decreased by a mean of 89.9 mg/24 h (-42.5%, p < 0.01 vs baseline and vs ramipril) in the imidapril-treated patients and by 63.2 mg/24 h (-29.6%, p < 0.05 vs baseline) in the ramipril-treated ones, with a

RIGHTSLINK

Table 2. Adverse events in the imidapril- and ramipril-treated patients.

Adverse events with withdrawn				Adverse events without withdrawn			
Ramipril		Imidapril		Ramipril		Imidapril	
No. of patients	Event	No. of patients	Event	No. of patients	Event	No. of patients	Event
5	Cough	1	Gastric discomfort	14	Cough	4	Cough
		1	Headache	3	Headache	2	Headache
		1	Gastric discomfort diarrhoea	2	Nausea	1	Nausea
				1	Diarrhoea	1	Diarrhoea
				1	Rash		

Table 3. Potassium and creatinine levels in the imidapril- and ramipril-treated patients.

	Imidapril (n = 88)		Ramipril (n = 88)		P
	Potassium (mEq/l)	Creatinine (mg/dl)	Potassium (mEq/l)	Creatinine (mg/dl)	
Baseline	4.41 ± 0.33	1.06 ± 0.19	4.36 ± 0.33	1.06 ± 0.16	ns
Week 2	4.48 ± 0.33	1.07 ± 0.19	4.43 ± 0.33	1.08 ± 0.16	ns
Week 4	4.54 ± 0.34	1.08 ± 0.19	4.51 ± 0.34	1.09 ± 0.16	ns
Week 6	4.50 ± 0.34	1.08 ± 0.19	4.46 ± 0.33	1.09 ± 0.16	ns
Week 12	4.45 ± 0.35	1.06 ± 0.19	4.41 ± 0.33	1.08 ± 0.16	ns
Week 24	4.41 ± 0.36	1.05 ± 0.19	4.35 ± 0.35	1.08 ± 0.16	ns
Week 26	4.41 ± 0.32	1.05 ± 0.19	4.36 ± 0.30	1.07 ± 0.16	ns

Data are expressed as mean ± SD.
ns = p not significant.

relative change from baseline between the two drugs of 13%, which was statistically significant ($p < 0.05$).

Two weeks after stopping therapy, UAE returned to pre-treatment values in both treatment groups, with no difference between them.

3.4 Biomarkers

Ang II plasma values significantly decreased from baseline with both drugs throughout the study. At week 24 end-point the mean decrease in Ang II was 14.6 pg/ml in the imidapril-treated patients and 7.9 pg/ml in the ramipril-treated patients (**Table 1**).

Bradykinin plasma levels were significantly increased by both agents, but such an increase was greater in the ramipril- than in the imidapril- treated patients at week 24 (+28.9 pg/ml vs +19.2 pg/ml), the difference between treatments being statistically significant ($p < 0.05$) (**Table 1**).

Both imidapril- and ramipril-based therapies produced a significant reduction in BNP mean levels (**Table 1**). At week 24 end-point BNP decreased by a mean of 13.7 pg/ml (-34.5%, $p < 0.05$ vs baseline) in the imidapril group and by a mean of 12.3 pg/ml (-32.3%, $p < 0.05$ vs baseline) in the ramipril group, with no significant difference between the two treatments.

3.5 Safety and tolerability

Both imidapril- and ramipril-based therapies were well tolerated. In the overall patient population, AEs were mild or moderate in intensity. The rate of AEs was 11% in the imidapril and 21% in the ramipril-treated patients. They included cough, which occurred in 4 patients (4.5%) in the imidapril group and in 14 patients (16%) in the ramipril group, headache (3 vs 2 patients, respectively), nausea (2 vs 1 patients), diarrhoea (1 patient in both group) and rash (1 patient with ramipril).(**Table 2**). Serum potassium and creatinine were unaffected by both treatments throughout the study (**Table 3**).

4. Discussion

The results of this study showed that in hypertensive patients with type 2 diabetes and microalbuminuria both imidapril- and ramipril-based therapies produced a significant reduction of UAE throughout the study period. However, the decrease in UAE associated with imidapril was always greater than that

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

RIGHTSLINK

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

associated with ramipril, although the difference between the two regimens became statistically significant from week 6 onwards and was maximal after 24 weeks of active treatment.

Mechanisms for such a different antiproteinuric effect are unclear. It cannot be related to a difference in BP decrease as the BP-lowering effect was similar in both treatment groups. This was true for both clinic and ambulatory BP, which is well known to provide a more complete evaluation of the antihypertensive efficacy of hypotensive agents [22]. In addition, in the present study we also assessed the effect of imidapril and ramipril on central BP which has been suggested to overcome some limitations that exist with peripheral BP measurements [18,23,24]. First, central BP seems to be a more accurate predictor of cardiovascular events than peripheral pressures [18]. Second, peripherally obtained systolic pressures, primarily measured from the brachial artery, do not accurately reflect pressure as recorded in the aorta and central arteries because of peripheral amplification [23]. Lastly, different antihypertensive agents have been shown to exert variable effects on central pressures despite similar reductions in brachial pressures, which might contribute to the distinct effects on cardiovascular outcomes obtained between different antihypertensive drugs [18,24]. In this study, imidapril and ramipril produced similar reduction in central pressures, with no significant difference between them.

Both imidapril and ramipril significantly reduced Ang II plasma levels from baseline. Interestingly, such a reduction was less marked, although not statistically, in the imidapril group. This might be due to a lesser Ang II escape phenomenon [25] under imidapril with consequent greater suppression of the RAAS with this ACE-I.

Since ACE is well known to act as kininase II to inactivate bradykinin with consequent accumulation of this substance [26], in both treatment groups bradykinin plasma values significantly increased from baseline. However, such an increase was significantly greater in the ramipril-treated patients. Since bradykinin is known to play a relevant role in the development of cough which is the main ACE-I-related side effect [27], the lesser increase in bradykinin concentration might at least partly explain the lower incidence of cough in the imidapril group, which confirms some previous observations [28,29]. Whether the different degree of bradykinin accumulation may be involved in the different effects of the two drugs on proteinuria remains to be clarified.

BNP levels, which have been suggested to predict cardiac and renal outcome in patients with diabetic nephropathy [30,31], were significantly reduced from baseline by both imidapril and ramipril, with no difference between the two treatments. This was probably expression of the reduced cardiac shear stress consequent to the BP lowering effect of the two drugs.

Based on our findings, we have no clear explanation for the observed different antiproteinuric effect of imidapril and ramipril. Since other authors have reported a limited efficacy of ramipril in reducing proteinuria in type 2 diabetic patients with nephropathy [9-11], we can only hypothesise that in this peculiar condition ramipril, unlike imidapril, may be less effective due to some intrinsic molecular characteristic. Differential tissue activity of the two different ACE inhibitors might be a possibility. In particular, different levels of tissue ACE inhibition at the kidney level might have played a role [32,33]. Another possibility might be a different degree of aldosterone suppression during long-term therapy with imidapril and ramipril. It has been shown that an increase in plasma aldosterone levels, so-called aldosterone escape, occurs in approximately 40% of type 2 diabetic patients with early nephropathy despite the use of ACE inhibitors, which might be responsible for the late escape of these patients from the antiproteinuric effect observed with these drugs [34]. We have no specific data on this issue, so that we can only hypothesise that imidapril might produce a more maintained decrease in plasma aldosterone levels as compared to ramipril with consequent less aldosterone breakthrough and more marked antiproteinuric effect.

The reduction in UAE certainly reflects an amelioration of the renal injury induced by increased mesangial and interstitial flux of protein. Whether this beneficial effect is coupled with an improvement in renal outcomes remains to be demonstrated.

5. Conclusion

The results of this study indicate that in hypertensive patients with type 2 diabetes and microalbuminuria, 24-week treatment with imidapril produced a greater antiproteinuric effect than ramipril, despite equivalent BP-lowering efficacy. Different intrinsic molecular characteristics might explain such a different effect, whose mechanisms remain unclear. Whether the observed different effect of imidapril and ramipril on UAE may have a different impact on the progression of renal disease in these patients remains to be demonstrated.

Acknowledgements

The authors wish to thank Mitsubishi Tanabe Pharma Corp. for providing editorial support in the development of this manuscript. The authors had full control over the content and writing of the paper.

Declaration of interest

The present study was carried out with the funds of the Department of Internal Medicine and Therapeutics-University of Pavia and the contribution of Mitsubishi Tanabe Pharma Corp. The authors declare to have no affiliation with, or financial involvement in, any organisation or entity with a direct financial interest in the subject matter or materials discussed in the manuscript.

RIGHTSLINK

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

Bibliography

Papers of special note have been highlighted as either of interest (•) or of considerable interest (••) to readers.

1. Parving HH, Mauer M, Ritz E. Diabetic nephropathy. In: Brenner BM, editor. Brenner and Rector's the kidney. 8th edition. WB Saunders; Boston: USA: 2008. p. 1265-98
2. Yamagishi S, Fukami K, Ueda S, Okuda S. Molecular mechanisms of diabetic nephropathy and its therapeutic intervention. Curr Drug Targets 2007;8:952-9
3. Lewis EJ, Hunsicker LG, Bain RP, Rohde RD. The effect of angiotensin-converting-enzyme inhibition on diabetic nephropathy. The Collaborative Study Group. N Engl J Med 1993;329:1456-62
4. The EUCLID Study Group. Randomised placebo-controlled trial of lisinopril in normotensive patients with insulin-dependent diabetes and normoalbuminuria or microalbuminuria. Lancet 1997;349:1787-92
5. Heart Outcomes Prevention Evaluation Study Investigators. Effects of ramipril on cardiovascular and microvascular outcomes in people with diabetes mellitus: results of the HOPE study and MICROHOPE Substudy. Lancet 2000;355:253-9
6. Brenner BM, Cooper ME, De Zeeuw D, et al. Effect of losartan on renal and cardiovascular outcomes in patients with type 2 diabetes and nephropathy. N Engl J Med 2001;345:861-9
7. Lewis EJ, Hunsicker LG, Clarke WR, et al. Renoprotective effect of the angiotensin-receptor antagonist irbesartan in patients with nephropathy due to type 2 diabetes. N Engl J Med 2001;345:851-60
8. Viberti G, Wheeldon NM. Microalbuminuria reduction with valsartan in patients with type 2 diabetes: a blood pressure independent effect. Circulation 2002;106:672-8
9. Rossing P, Hommel E, Smidt UM, Parving HH. Reduction in albuminuria predicts a beneficial effect on diminishing the progression of human diabetic nephropathy during antihypertensive treatment. Diabetologia 1994;37:511-16
10. Lambers Heerspink HJ, Brinkman JW, Bakker SJ, et al. Update on microalbuminuria as a biomarker in renal and cardiovascular disease. Curr Opin Nephrol Hypertens 2006;15:631-6
11. Trevisan R, Tiengo A. Effect of low-dose ramipril on microalbuminuria in normotensive or mild hypertensive non-insulin dependent diabetic patients. Am J Hypertens 1995;8:876-83
12. Schnack CH, Hoffman W, Hopmeier P, Schernthauer G. Renal and metabolic effects of 1-year treatment with ramipril or atenolol on NIDDM patients with microalbuminuria. Diabetologia 1996;39:1611-16
13. Marre M, Lievre M, Chatellier G, et al. Effects of low dose ramipril on cardiovascular and renal outcomes in patients with type 2 diabetes and raised excretion of urinary albumin: randomized, double-blind, placebo-controlled trial (the DIABHYCAR study). BMJ 2004;328:495

• **This large study demonstrated that low dose ramipril had no significant beneficial effects over placebo in type 2 diabetic patients with both incipient and overt nephropathy after 4 years of follow-up.**

14. Oda H, Kachi R, Ohashi H. Clinical usefulness of imidapril hydrochloride in diabetic nephropathy with hypertension. Progr Med 1995;15:2093-8
15. Umeda M. Clinical effect of ACE-inhibitor imidapril on proteinuria excretion in hypertensive patients with type 2 diabetes. Jpn J Clin Exp Med 1999;76:1001-7

•• **This study first demonstrated that imidapril was effective in reducing urinary albumin excretion in hypertensive type 2 diabetics with microalbuminuria.**

16. Groppelli A, Omboni S, Ravogli A, et al. Validation of the SpaceLabs 90202 and 90207 devices for ambulatory blood pressure monitoring by comparison with intra-arterial resting and ambulatory measurements. J Hypertens Suppl 1991;9:S336
17. Parati G, Bosi S, Castellano M. Guidelines for 24-h non-invasive ambulatory blood pressure monitoring: report from a working group of the Italian Society of Hypertension. High Blood Press 1995;4:168-74
18. Williams B, Lacy PS, Thom SM, et al. Differential impact of blood pressure lowering drugs on central aortic pressure and clinical outcomes: principal results of the Condruit Artery Function Evaluation (CAFE) study. Circulation 2006;113:1213-25
19. Schmieder RE, Langenfeld MR, Friedrich A, et al. Angiotensin II related to sodium2 excretion modulates left ventricular structure in human essential hypertension. Circulation 1996;94(6):1304-9
20. Billings FT IV, Balaguer JM, C Y, et al. Comparative effects of angiotensin receptor blockade and ACE inhibition on the fibrinolytic and inflammatory responses to cardiopulmonary bypass. Clin Pharmacol Ther 2012;91(6):1065-73
21. Rawlins ML, Owen WE, Roberts WL. Performance characteristics of four automated natriuretic peptide assays. Am J Clin Pathol 2005;123(3):439-45
22. Mancia G, Omboni S, Ravogli A, et al. Ambulatory blood pressure monitoring in the evaluation of antihypertensive treatment: additional information from a large data base. Blood Press 1995;4:148-56
23. Nelson MR, Stepanek J, Cevette M, et al. Noninvasive measurement of central vascular pressures with arterial tonometry: clinical revival of the pulse pressure waveform? Mayo Clin Proc 2010;85:460-72
24. Morgan T, Lauri J, Bertram D, Anderson A. Effect of different antihypertensive drug classes on central aortic pressure. Am J Hypertens 2004;17:118-23
25. Hollenberg NK, Fisher ND, Price DA. Pathways for angiotensin II generation in intact human tissue: evidence from comparative pharmacological interruption of the renin system. Hypertension 1998;32:387-92
26. Skidgel RA. Bradykinin-degading enzymes: structure, function and potential roles in cardiovascular pharmacology. J Cardiovasc Pharmacol 1992;20(Suppl 9):S4-9

RIGHTSLINK

27. Fox A, Lalloo U, Belvisi M, et al. Bradykinin-evoked sensitization of airway sensory nerves:a mechanism for ACE-inhibitor cough. Nat Med 1996;2:814-17

28. Sasaguri M, Ideishi M, Kinoshita A. Differential inhibition of bradykinin hydrolisis by four ACE inhibitors: a possible explanation for differences in induced coughing. Hypertens Res 1994;17:253-8

- **This study suggested that the different effect of ACE-inhibitors in cough inducing may be related to differential inhibition of bradykinin hydrolisis.**

29. Boonyapisit W, Tresukosol D. Comparison of the incidence of imidapril and enalapril induced cough. J Med Assoc Thai 2010;93(Suppl 1):S48-53

30. Soleimani A, Nasiri O, Nikoueinejad H, et al. Prognostic value of B-type natriuretic peptide for assessment of left ventricular function in patients with chronic kidney disease. Iran J Kidney Dis 2011;5:242-7

31. Seki N, Nishimura M, Matsumoto T, et al. Relationship between BNP level and renal function in diabetic nephropathy with microalbuminuria. J Diabetes Complications 2013;27:92-7

32. Taniguchi M, Kim S, Zhan Y, Iwao H. Role of intrarenal converting-enzyme in nephropathy of type II diabetic rats. Hypertens Res 2002;25:287-94

33. Hosova K, Ishimitsu T. Protection of cardiovascular system by imidapril,a versatile angiotensin-converting enzyme inhibitor. Cardiovasc Drug Rev 2002;20:93-110

34. Sato A, Hayashi K, Naruse M, et al. Effectiveness of aldosterone blockade in patients with diabetic nephropathy. Hypertension 2003;41:64-8

- **This study suggested that aldosterone escape may occur in long-term treatment with an ACE-inhibitor and may reduce its antiproteinuric effect in patients with diabetic nephropathy.**

Affiliation

Roberto Fogari, Amedeo Mugellini, Annalisa Zoppi, Roberta Gualtierotti, Pierangelo Lazzari, Giuseppe Derosa & Angela D'Angelo
Centro per l'Ipertensione e la Fisiopatologia Cardiovascolare, Department of Internal Medicine and Therapeutics, Piazzale Golgi 19, 27100 Pavia, Italy
Tel: +390382526217;
Fax: +390382526259,
E-mail: r.fogari@unipv.it.

Expert Opin. Pharmacother. Downloaded from informahealthcare.com by QUT Queensland University of Tech on 11/02/14
For personal use only.

RIGHTSLINK